PREVENTATIVE MAINTENANCE CHECKLIST

Enclosed Cab	Underground Haulage Truck	Tier 4	Dump

250	Hour

DATE: _____

UNIT NUMBER: _____

TECHNICIANS: _____

HOURS: _____

WORK ORDER: _____

***** PM TO BE COMPLETED ONCE THE UNIT IS WASHED *****

Note: All specifications, procedures, and parts not identified in this PM can be found in the OEM manuals

PRE-WORK SAFETY TASKS

MANDATORY PPE

LOCKOUT / TAGOUT – Ensure all procedures are followed to eliminate stored energy

Before working on brake system or hydraulic system drain accumulator pressure

Ensure deck, walkways and steps are free from foreign material such as oil and debris and that traction surface is in place and in good condition

Read and understand all warning plates on machine before operating, lubricating or repairing machine

To avoid burns, be alert for hot parts on machines, which have just been stopped and have hot fluids in lines, tubes and compartments.

Battery electrolyte contains acid – avoid contact with skin or eyes.

Perform all running checks in an open area free of obstruction and personnel

Ensure the box is locked in the raised position and the steering lock is in place before working on the machine

Ensure the box is free of muck before working on the machine

If working on air system brakes, bleed down air system

Ensure moldboard and ripper are lowered to the ground before servicing equipment

Refer to manufacturer's operation and maintenance manuals for equipment specifications, fluid capacities, operating procedures and maintenance procedures

Be sure the area around the machine is clear of personnel and obstructions while performing running checks

FOR MORE COMPLETE LIST OF SAFETY MESSAGES, REFER TO THE OPERATION AND MAINTENANCE MANUAL FOR THIS MACHINE

a = OK, b = ADJUST THING AND REPAIRED, c = ADJUST THING NOT REPAIRED, D = N/A
⚠ = TAKE MACHINE OUT OF SERVICE ⚠ = CONSTRUCT PROTECTIVE PROOF

		A	B	C	D	⚠
PRE-ISOLATION TASKS AND INITIAL WALK AROUND						
1	Wash units (remove bolt) pan before washing engine area!					
2	Pre-op book - check unit has been & crosses book in current & remount					
3	Pre-op book comments - review					
4	Check condition of guard walls, ensure they all unbolted and lock in place					
5	Check "STORED ENERGY" labels, make sure they are in place and legible					
6	Ensure key lock is equipped with two detail checks, that they are in good condition, and that the storage mounts on the machine are in good shape					
7	Check condition & operation across doors, lids & prop rods, verify that all weather surfaces are in good shape and the steps and grab handles are safety.					
8	Check Front & rear frames for cracks & expect damage					⚠
9	Check the frame rub bars for excess wear					
OPERATOR STATION						
10	Check proper operation of the dash alarm beeper, vehicle horn and backup alarm					⚠
11	Test the operation of the neutral lockout key to start machine while in gear!					⚠
12	Test the operation of the function lockout					⚠
13	Gauges, dash lights, marker lights, running lights - check condition & operation					
14	Operator control labels - ensure they are in place & legible					
15	Check proper operation of the seat door gas spring assembly and ensure the hinges are in good condition					
16	Labels - inspect labelling on with CIPR, warning, operator labels!					
17	Brake test attenuation alarms - check if it is in place					
18	Brake pedal - check for smooth operation, wear of pivot & remove any obstructions on floor					⚠
19	Ensure ABS is functioning properly					⚠
20	Service tanks holding capacity - test					⚠
21	Park tanks holding capacity - test					⚠
22	Throttle - check for smooth operation & remove any obstructions on floor					⚠
23	Decelerator - check for smooth operation & remove any obstructions on floor					⚠
24	Surface TCP download, provide download file to Flameon / Reliability					
25	Seat and armrests - check overall condition and ensure the height and suspension adjusters are working					
26	Seatbelt - check condition & expiry date					⚠
27	ROPS / FOPS - check for damage & loose mounting bolts					
28	Hydraulic & transmission control levers - check for free movement & looseness					⚠
29	Check condition of the door, hinges, latches, latching strap, and door seals					
30	Oil leaks in cab & under door panel - check for leaks					
31	Check condition of windshield wipers and verify the wipers and washers are working					
32	Windows - check condition and ensure the emergency egress tab is in place					⚠
33	Check operation of the heating / air conditioning system including the fan speed and temperature switches, ensure the evaporator drains are clear and the directional vents are pressure and in good condition.					
REVERSING CHECKS						
34	Engine breather tube - check for excessive smoke or leaking engine oil					
35	Transmission - check for oil leaks					
36	Forward / reverse - operate only, listen for noise and ensure that all gears are in good condition					⚠
37	Surface torque converter drive and shaft load					⚠
38	Verify operation of the horn lockup					
39	Verify all hydraulic functions are working, including float					
40	Check operation of the payload monitoring system, calibrate system					
41	Ensure the tow up gear lockout is working correctly					
42	Check for play in the articulation pins, ensure that all of the bolts are tight					⚠
43	Steering stop - check clearance both caps to avoid steering cylinders bottoming out, adjust steering stops and restrictor valves if necessary.					
44	Verify secondary (emergency) steering is operational					
45	Check for play (including side to side play) in the steering cylinder pins and bushings. Ensure pin retainers are tight and in place					
46	Check for play (including side to side play) in the lift cylinder pins and bushings. Ensure pin retainers are tight and in place					
47	Check the operation of the backup camera, ensure the camera is aimed properly. Position pylon 1 under camera, pylon 2 at 6' (2m) directly in line with 1, and pylon 3 directly in line with 1 and 2 as far away as possible. Position pylons 4 and 5 at 3' (1m) directly in line with 1 and 2 at the camera extremities respectively and halfway between pylons 1 and 2 Position pylons 6 and 7 directly 6' (2m) in line left and right respectively. Pylon 8 Note: Position as shown in illustration. Camera view (illustration 2): Pylons 1, 2, and 3 must be in a straight line as per straight line in camera view. Pylons 4, and 5 must be the same distance to the outside of the machine as per arrows in camera view; Pylons 6 and 7 must not be seen by the camera					
ISOLATION						
48	Lift machine and place on jack stands (as needed)					
49	Bleed hydraulic and brake pressures					
50	Install steering lock and tow lock					
51	Tag and lockout machine					
52	Barricade working area					
53	Ensure cabin tilt function is working properly and that the cylinder is in good condition					
54	Install cabin tilt lock					
OIL SAMPLES + OIL AND FILTER CHANGES						
55	Take engine oil sample, check oil level					
56	Replace water separator & fuel filter					
57	Replace engine fuel filter					
58	Engine coolant - take sample, check coolant level					
59	Transmission oil - take sample, check oil level					
60	Quarter oil - take sample, check oil level					
61	Replace Crankcase/ Fumes disposal filter (DCV)					
62	Clean DPF filter screen					
63	Hydraulic system - take sample, check oil level					
64	Front and rear differential oil - take sample, check oil level					
65	Take oil sample, check final drive levels & check for leaks - top up if necessary					
66	Take oil sample, check swing drive oil level					
67	Take samples, check all final drive levels & wheel ends for leaks - top up if necessary					
68	Ensure proper labelling and packaging of samples to be sent out for analysis					
69	Replace primary air filter					
70	Replace cab air filter					
71	Replace recirculating air (A/C) filter					
ENGINE COMPARTMENT						
72	ANTIFREEZE - check condition & tightness of mount bolts, inspect fins for damage and certify the fins are not plugged and the rotors are not leaking					
73	AFTERCOOLER - check condition & tightness of mount bolts, inspect fins for damage and certify the fins are not plugged and the rotors are not leaking					
74	HYDRAULIC COOLER - check condition & tightness of mount bolts, inspect fins for damage and certify the fins are not plugged and the rotors are not leaking					
75	ENGINE OIL COOLER - check condition & tightness of mount bolts, inspect fins for damage and certify the fins are not plugged and the rotors are not leaking					
76	AIR CONDITIONING CONDENSER - check condition & tightness of mount bolts, inspect fins for damage and certify the fins are not plugged and the rotors are not leaking, and ensure fans are secured to the condenser body and working					
77	INTAKE - check condition of manell system, including filter housing and precleaner, piping to and from engine and aftercooler, and the intake manifold for cracks, damage, loose clamps, leaks, bolts, nuts, etc and rubber hose for degradation, and all components					⚠
78	EXHAUST - check condition of manell system including piping, clamps, gaskets and exhaust stack or piping					
79	BATTERIES - check all terminals and posts are free from corrosion and securely fastened					
80	Check belt tensioner and condition					⚠
81	Check condition of all hoses for leaks					
82	Check condition of engine covers and check all guards are in place & secure					
83	Oil & fuel leaks - check engine bay for leaks & cleanliness					
84	Hydraulic, transmission and breaks/steering pumps - check for leaks					
TIRES						
85	Check tires for large sections missing from the face, face peeling, large cuts and gouges, sidewall damage (depth damage), sidewall cuts (cuts, exposed), or any tube bulging					⚠
86	Check condition of rims for cracks, bends, missing or loose studs / nuts, elongated bolt holes, damage to rims lock ring					⚠
87	Please record serial numbers from the tires: 2A 1A 1B 2B 2C 3C 3D 4D 4C 3B 3A					
88	Measure inside & outside tire depth, record lowest reading (in /32") 2A 1A 1B 2B 2C 3C 3D 4D 4C 3B 3A					★ ★ ★ ★ ★ ★ ★ ★
89	Measure diameter with tape measure across center of tire (ensure there is no more than 2% difference side to side and 8% difference front to back) 2A 1A 1B 2B 2C 3C 3D 4D 4C 3B 3A					★ ★ ★ ★ ★ ★ ★ ★
90	Check inflation pressure in tires: 2A 1A 1B 2B 2C 3C 3D 4D 4C 3B 3A					★ ★ ★ ★ ★ ★ ★ ★
POWERTRAIN						
91	Check condition of transmission mounts & torque mounting bolts					⚠
92	Inspect all universal joints for wear and play; check & tighten bolts					⚠
93	Check tightness of oscillating axle bolts, check for play in the oscillation bushings					
94	Check front, center, and rear axle mounting frame pivot pins and bushings for play, ensure they are taking grease					⚠
95	Check front center, and rear, stabilizer links for excess play or damage					⚠
96	Check suspension cylinders for leaks or damage, measure suspension height to ensure the cylinder is still in good condition					⚠
97	Check front, center and rear axle mounting bolts using hammer (tapping) to identify any loose bolts					⚠
98	Check torque / transmission compartment for any leaks, check hoses for wear, damage and proper routing, check electrical harnesses for proper routing and any abrasion					
99	Check condition of ground strap in the articulation hitch					
DUMP BODY / EJECTOR BOX						
100	Check box for wear and cracks					
101	Check for excess wear or gaps on box stem support pads					
102	Ensure grease hoses are in good condition and all pins are taking grease					⚠
103	Inspect condition dump cylinders					
HYDRAULICS AND HYDRAULIC TANK COMPARTMENT						
104	Check condition & routing of all hoses in hydraulic tank bay					
105	Check condition & routing of all hoses in articulation area					
106	Check condition & routing of all hoses in rear frame area					
107	Check condition & installation of hydraulic sensor					
108	Check hydraulic tank condition & mounting, check for leaks					
109	Check all valves for leaks					
DRAINING						
110	Check level of the autogreaser reservoir					
111	Check for proper operation of grease pump					⚠
112	Check for leaks at all lines & connections on auto lube system, if equipped					⚠
113	Ensure for bearing on engine					
114	Check for proper grease distribution at all auto lube points					
115	Ensure all universal joints, ship joints & pillow block					
116	Ensure all cover and wear hinges					
POST WORK TASKS						
117	Replace all fluid levels and verify all drains and level plugs are tight					
118	Make sure every closure, access panel, cover, and door is properly closed and locked					
119	Clean work area of debris, old parts, excess / unused materials and lubricants, refill up hoses and cards, return chemicals to flammable cabinet.					
120	Return equipment period or so inspected to a safe operating condition					
121	Ensure work orders are submitted to planning for all fixed corrective issues to be scheduled at a later date					
122	Ensure all reports and paperwork is signed off, dated and returned to Supervisor / Planning					
123	Walk around equipment to ensure that all foreign materials are cleaned up & all tools/equipment are accounted for, ensure that all tools are in proper condition					
124	Complete paperwork - ensure all work orders are filled in properly, found work scope sheets are filled out & handed into supervisor for review.					
125	Complete parts & core returns					
126	Clean up old Pillows / oils and dispose of properly					
RE-ISOLATION						
127	Remove lock pin and tower cable, ensure OK to operate					
128	Remove padlock & danger tags & re-energize equipment					
129	Remove articulation lock, tow up lock, tower equipment to the ground and remove barricade.					
START UP						
130	Start unit and inspect for any fluid leaks					
131	Verify all fluids levels					
132	Verify operations when work is completed					

SIGN OFF

TECHNICIAN NAME:

TECHNICIAN SIGNATURE:

SUPERVISOR NAME:

SUPERVISOR SIGNATURE:

DATE:

DEFFICIENCY LIST
CATERPILLAR SCOOP (LHD)
HR - PM

Step No.	Defficiency	Corrective Action Required	Work Request No.	Comp

PM FEEDBACK FORM

HAVE YOU FOUND ANY PARTS DISCREPANCIES?			
STOCK #	***INCORRECT PART #***	***CORRECT PART #***	***DESCRIPTION OF PART***

HAVE YOU IDENTIFIED ANY MAINTENANCE TASK TO BE UPDATED, ADDITION OR DELETED?	
TASK DESCRIPTION:	***UPDATE / ADD / DELETE (circle one)***